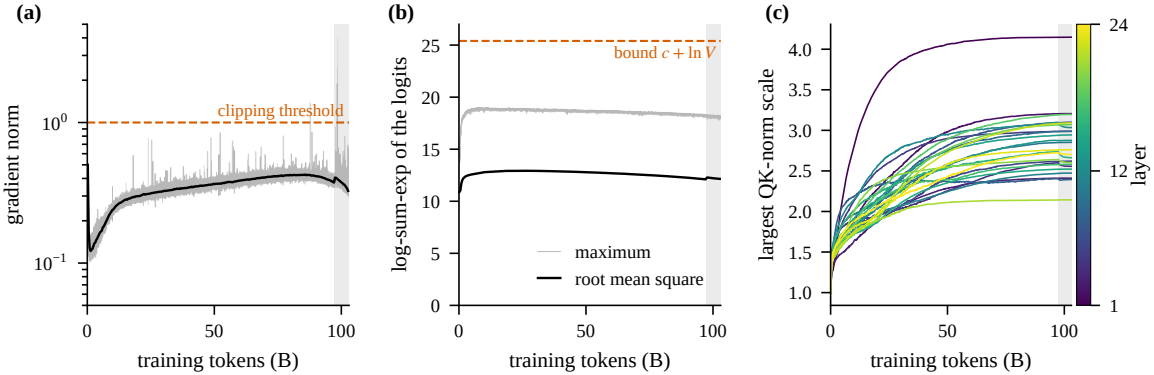

(a)
gradient norm
clipping threshold
10^0
10^{-1}
0
50
100
training tokens (B)
(b)
log-sum-exp of the logits
bound $c + \ln V$
maximum
root mean square
0
5
10
15
20
25
0
50
100
training tokens (B)
(c)
largest QK-norm scale
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
50
100
training tokens (B)
24
12
1
layer